Chapter 15: Wavelets

(i) Fourier transform

$$F(u)=\frac{1}{N}\sum_{x=0}^{N-1} f(x)\exp(\frac{-j2\pi xu}{N}),\quad u=0,1,\cdots,N-1$$

requires that the entire input signal

$$f(0)\,f(1)\ \cdots\cdots\ f(N-1)$$

be readily available for calculating every $F(u)$

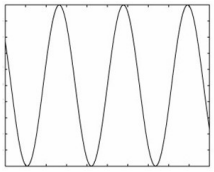
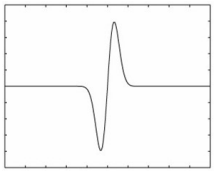
(ii) Fourier spectrum $|F(u)|$, $u=0,1,\cdots,N-1$ provides all the frequencies present in a signal but does not tell where they are present.

Windowed Fourier transform suffers from

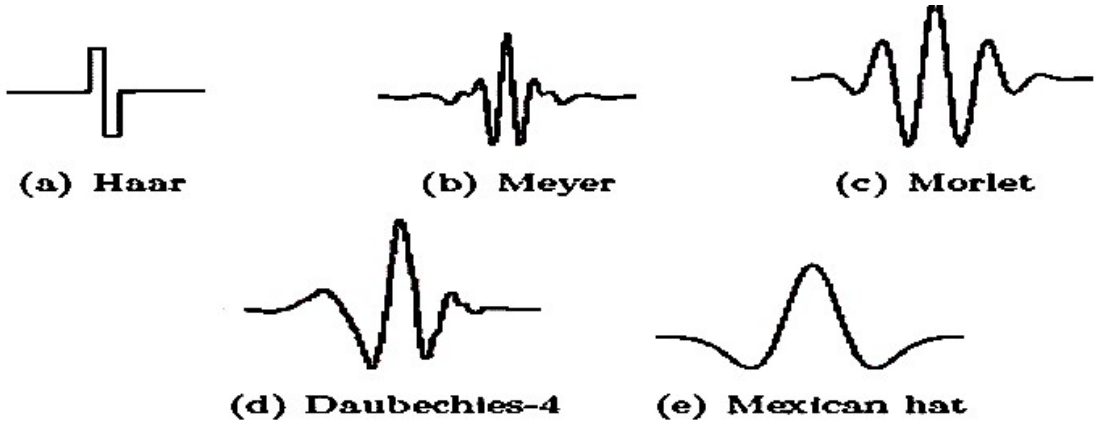
Small range – poor frequency resolution

Large range – poor localization

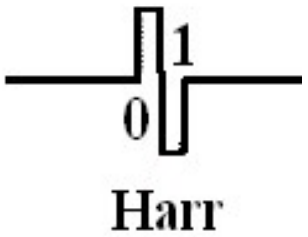
Wavelet: wave that is only nonzero in a small region

Wave   Wavelet

Types of wavelets:

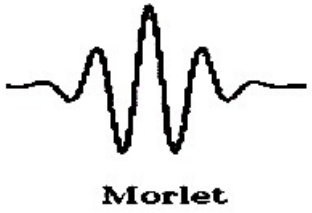


Haar:



$$\psi(x) = \begin{cases} 1 & \text{if } 0 < x < 1/2 \\ -1 & \text{if } 1/2 \le x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Morlet:



$$\psi(x) = e^{-x^2} \sin x,$$

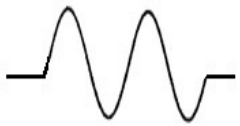
Mexican hat:



DOG, LOG

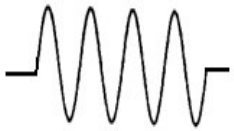
$$\text{LOG: } \psi(x) = \frac{2}{\sqrt{3\sigma}\pi^{1/4}}\left(1 - \frac{x^2}{\sigma^2}\right)e^{-x^2/(2\sigma^2)}$$

○ Operations on wavelet: $\psi(x)$ e.g.,



(a) Dilation:

i) Squashing $\psi(2x)$



ii) Expanding $\psi(x/2)$



(b) Translation:

i) Shift to the right

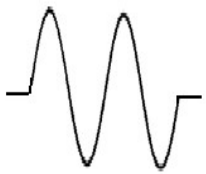
$\psi(x-2)$

ii) Shift to the left

$\psi(x+2)$

(c) Magnitude change:

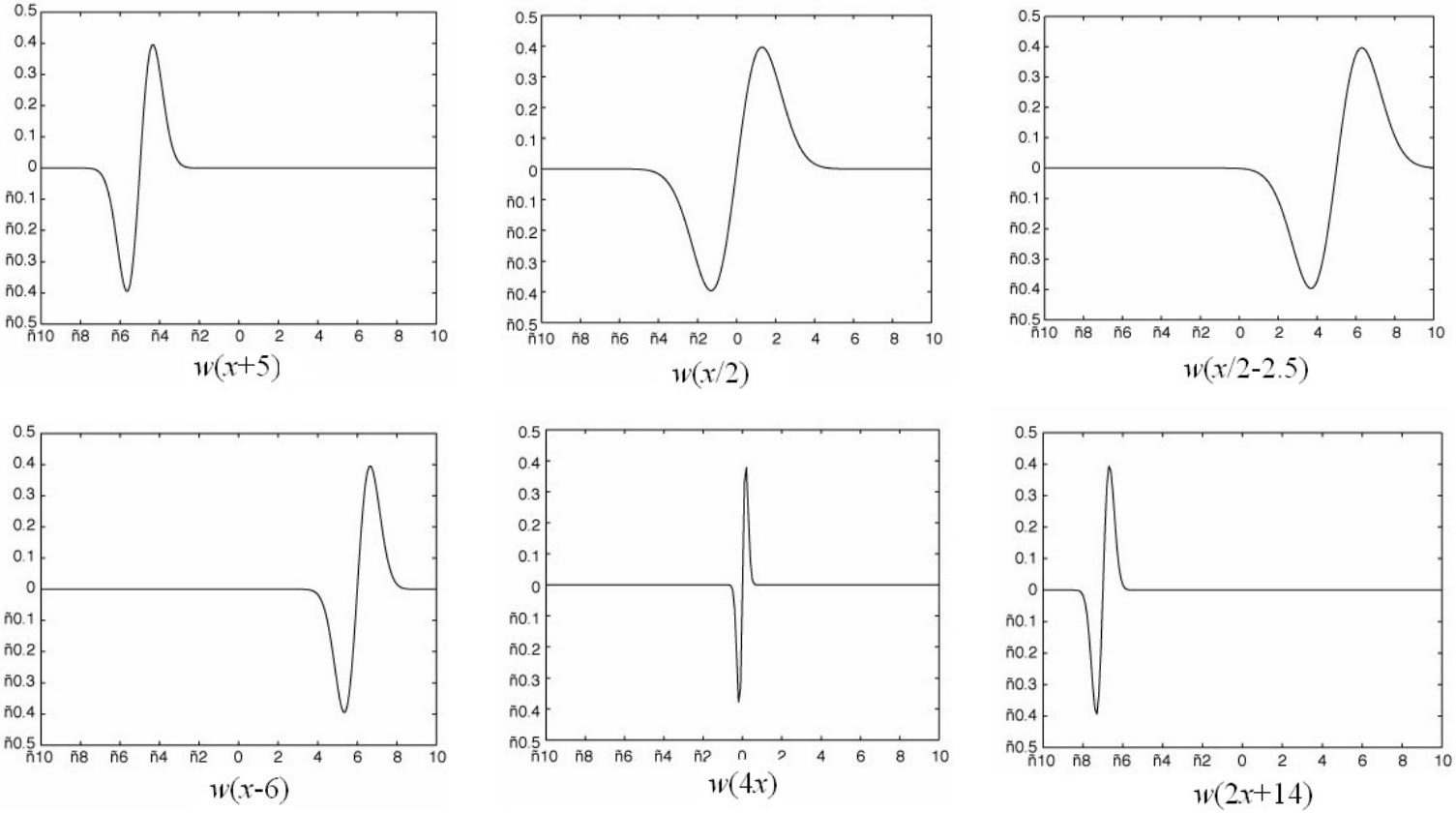
i) Amplification

$2\psi(x)$ 

ii) Minification

$1/2\psi(x)$ 

- Any function can be expressed as a sum of wavelets of the form $a_i \psi(b_i x + c_i)$, where a_i, b_i, c_i govern the amplitude, frequency and translation of wavelet i.



Discrete Wavelet Transform:

Approximation coefficients (*cA*)

$$W_\phi(i,j) = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} f(x)\phi_{ij}(x), \quad \phi_{ij}(x): \textbf{ scaling functions}$$

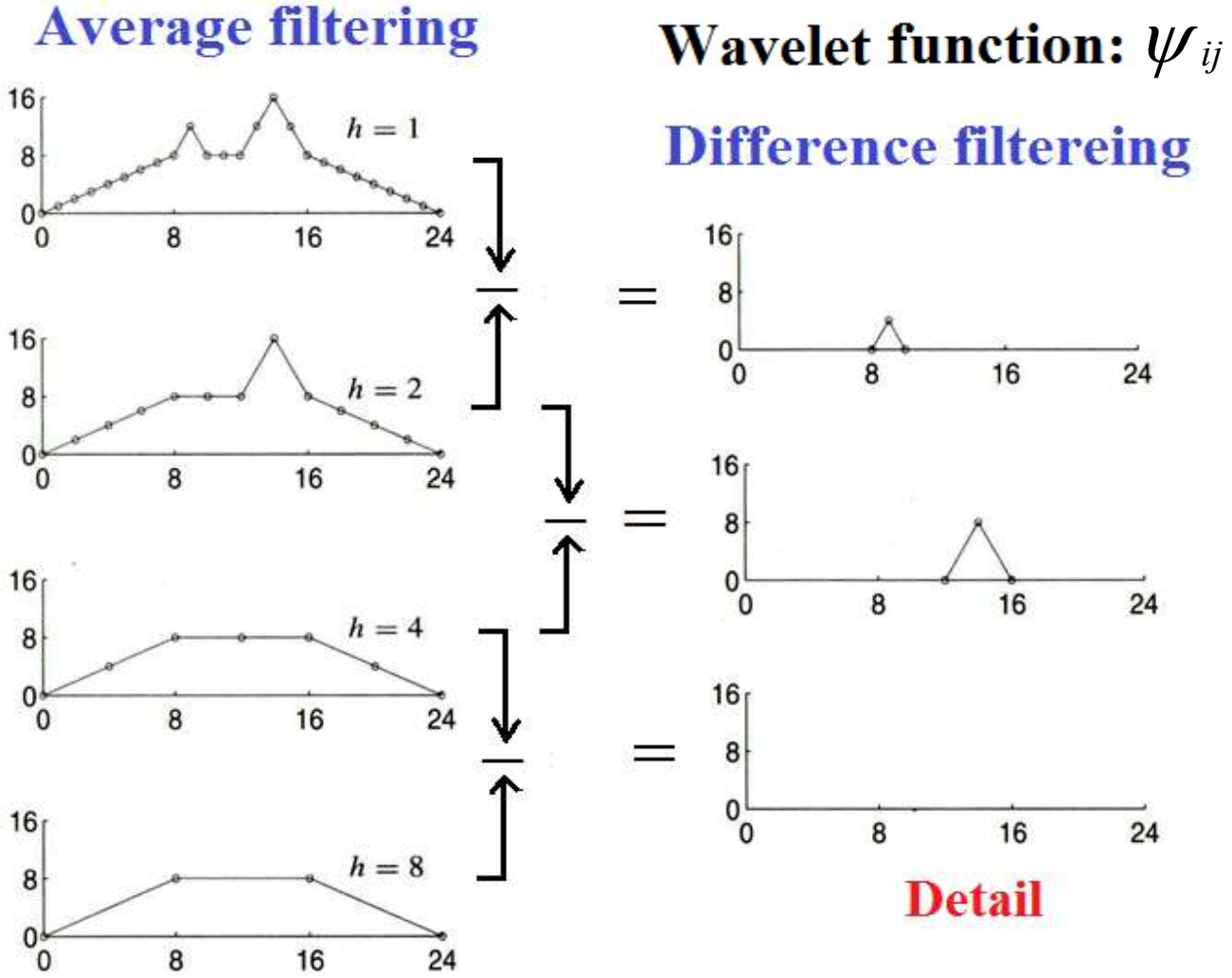
Detail coefficients (*cD*)

$$W_\psi(i,j) = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} f(x)\psi_{ij}(x), \quad \psi_{ij}(x): \textbf{ wavelet functions}$$

$\phi_{ij}(x)$: creates a series of approximations of $f(x)$.

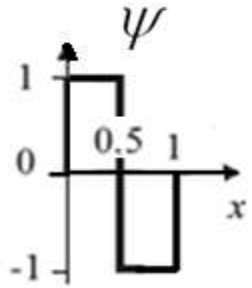
$\psi_{ij}(x)$: encodes differences between sucessive approximations.

e.g., **Scaling function:** ϕ_{ij}



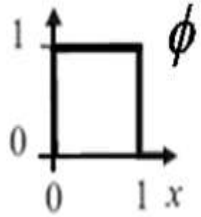
Example:

Haar wavelets $\psi(x) = \begin{cases} 1 & 0 \le x < 1/2 \\ -1 & 1/2 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$

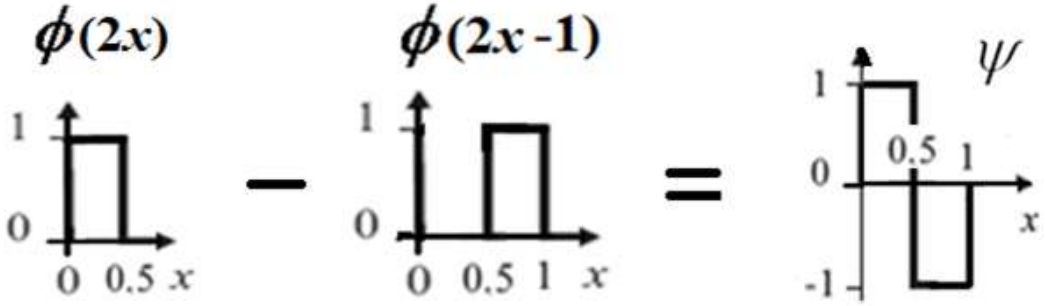


$\psi(x)$ can be written as $\psi(x) = \phi(2x) - \phi(2x-1)$,

where $\phi(x) = \begin{cases} 1 & 0 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$:



Haar scaling function $\phi(x)$



$\psi(x)$ and $\phi(x)$: **mother functions**.

Dilations, translations, and magnitude changes of $\psi(x)$ and $\phi(x)$, i.e., $\phi_{ij}(x), \psi_{ij}(x)$, are based on powers of 2.

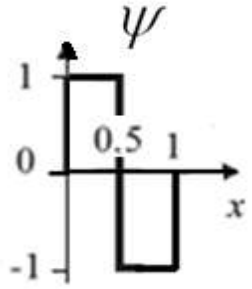
$$\phi_{ij}(x) = 2^{i/2}\phi(2^i x - j) = 2^{i/2}\phi(2^i(x - j/2^i))$$

$$\psi_{ij}(x) = 2^{i/2}\psi(2^i x - j) = 2^{i/2}\psi(2^i(x - j/2^i)),$$

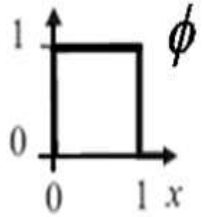
$$j = 0, \ldots, 2^i - 1$$

$\phi_{ij}(x), \psi_{ij}(x)$: **basis functions** of Haar wavelet transform

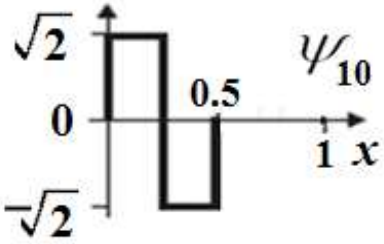
Example: Haar wavelets $\psi(x)$:

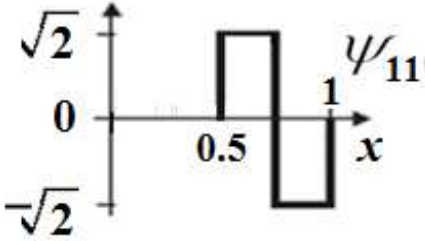


Haar scaling function $\phi(x)$:

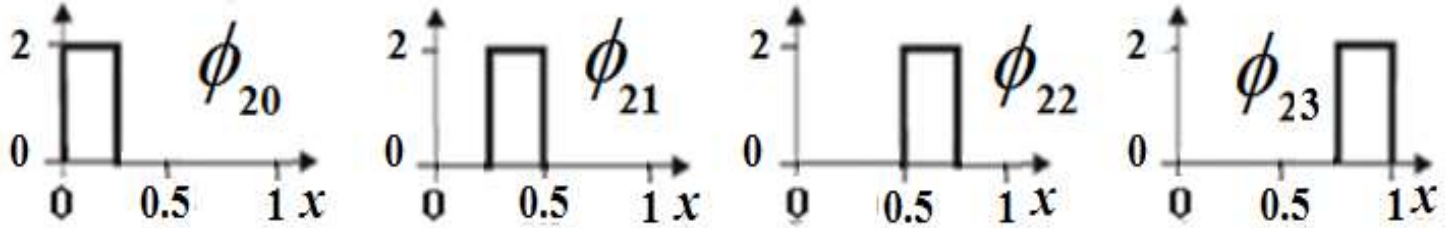


$$\psi_{ij}(x) = 2^{i/2}\psi(2^i(x - j/2^i))$$





$$\phi_{ij}(x) = 2^{i/2}\phi(2^i(x - j/2^i))$$



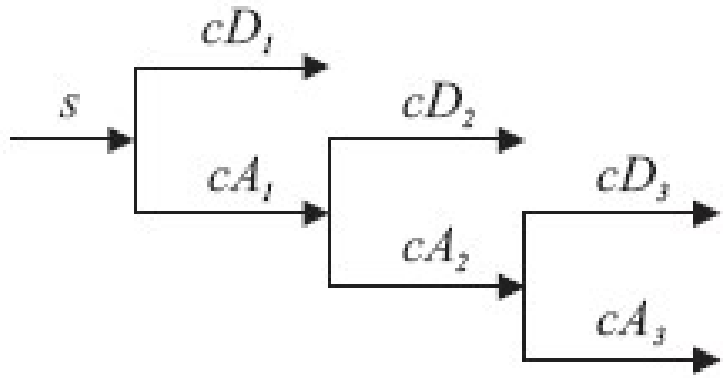
Inverse discrete wavelet transform:

$$f(x) = \frac{1}{\sqrt{N}} \sum_j W_\phi(i,j)\phi_{ij}(x) + \frac{1}{\sqrt{N}} \sum_j W_\psi(i,j)\psi_{ij}(x)$$

Fast Wavelet Transform

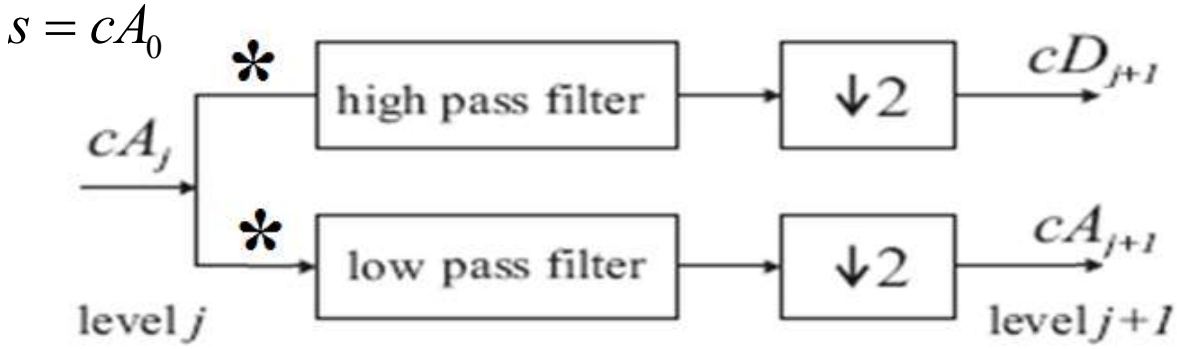
Discrete signal $s(n),\ n = 0,\ 1,\ \cdots,\ N-1$

to be transformed into wavelet coefficients



cD: detailed coef.

cA: approximate coef.



$*$: convolution $\quad \downarrow 2$: downsampling by 2

$\downarrow 2$: discard every other convolved value

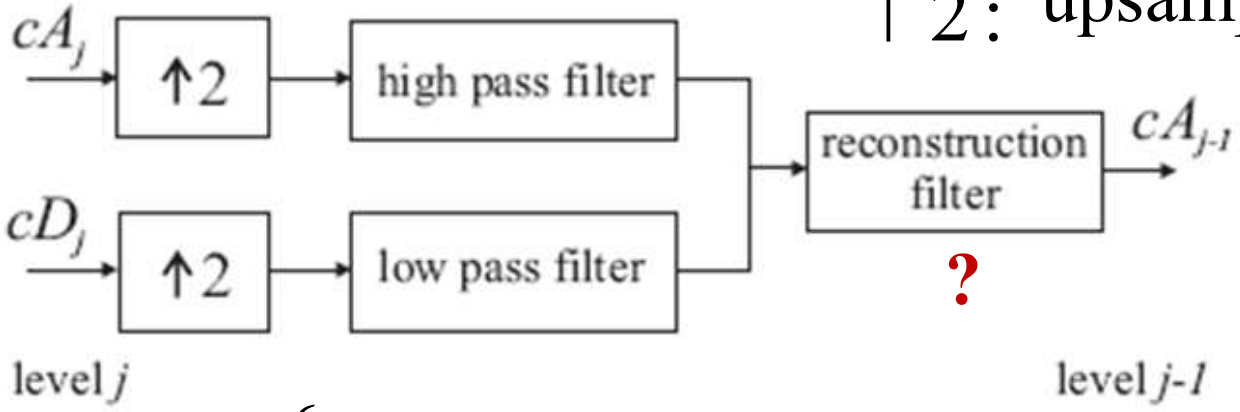
e.g., Giving sequence $y(n),\ n = 0,1,2,\cdots$

its downsampled sequence is

$$y_{\downarrow 2}(n) = y(2n),\ n = 0,1,2,\cdots$$

Inverse Wavelet Transform

$\uparrow 2$: upsampling by 2



$$y_{\uparrow 2}(n) = \begin{cases} y(n/2) & n\text{: even} \\ 0 & \text{otherwise} \end{cases},\ n = 0,1,2,\cdots$$

- Wavelet transforms work by taking low pass filtering (LPF) + and high pass filtering (HPF) – of input data

  e.g., Input data: (a, b)

  LPF (+): $s = a + b$; HPF (–): $d = a - b$

  Wavelet coefficients: (s, d).

- Inverse wavelet transforms work by taking LPF (+) and HPF (–) of wavelet coefficients, e.g., (s, d)

  LPF(+) : $s + d = (a+b) + (a-b) = 2a$,

  HPF(–) : $s - d = (a+b) - (a-b) = 2b$,

  Reconstructipn filter (/2)

  Inverse wavelet coefficients: (a, b).

- Example: Input data (14, 22)

Wavelet transform:

$s = 14 + 22 = 36,\ d = 14 - 22 = -8$

Transform result (36, –8).

Inverse wavelet transform:

Input data (36 –8)

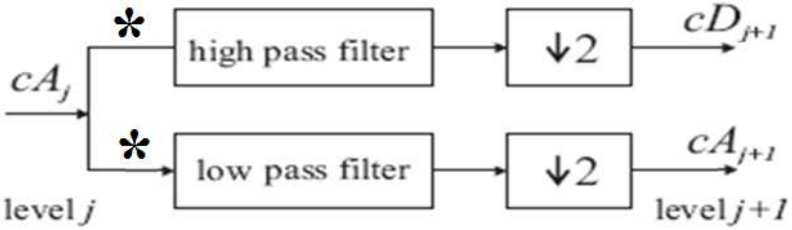
$s + d = 36 + (-8) = 28,$

$s - d = 36 - (-8) = 44$

(28, 44) / 2 = (14, 22)

Inverse transform result (14, 22).

◦ Example: Input data: [71 67 24 26 36 32 14 18]



Wavelet transform:

$s1 = [(71+67)\ \ (24+26)\ \ (36+32)\ \ (14+18)]$

$= [138\ \ 50\ \ 68\ \ 32]$ ***(cA)***

$d1 = [(71–67)\ \ (24–26)\ \ (36–32)\ \ (14–18)]$

$= [4\ \ –2\ \ 4\ \ –4]$ ***(cD)***

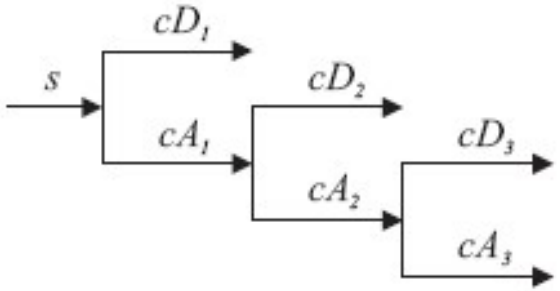
Transform result at 1 scale:

$v1 = [s1\ \ \ d1] = [138\ \ 50\ \ 68\ \ 32\ \ 4\ \ –2\ \ 4\ \ –4]$

(cA) ***(cD)***

In matrix form:

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 71 \\ 67 \\ 24 \\ 26 \\ 36 \\ 32 \\ 14 \\ 18 \end{bmatrix} = \begin{bmatrix} 138 \\ 50 \\ 68 \\ 32 \\ 4 \\ -2 \\ 4 \\ -4 \end{bmatrix}$$



(*cA*): [138 50 68 32]

$s2 = [(138+50) \;\; (68+32)] = [188 \;\; 100]$

$d2 = [(138–50) \;\; (68–32)] = [88 \;\; 36]$

Transform result at 2 scale:

$v2 = [s2 \;\; d2] = [188 \;\; 100 \;\; 88 \;\; 36]$

In matrix form:
$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 138 \\ 50 \\ 68 \\ 32 \end{bmatrix} = \begin{bmatrix} 188 \\ 100 \\ 88 \\ 36 \end{bmatrix}$$

$s3 = [(188+100)] = [288],\ \ d3 = [188–100] = [88]$

Transform result at 3 scale:

$v3 = [s3\ \ d3] = [288\ \ 88]$

In matrix form: $\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 188 \\ 100 \end{bmatrix} = \begin{bmatrix} 288 \\ 88 \end{bmatrix}$

Inverse wavelet transform:

Input data: $[s3\ \ d3\ \ d2\ \ d1] =$

$[288\ \ 88\ \ 88\ \ 36\ \ 4\ \ -2\ \ 4\ \ -4]$

$$[(288+88)/2 \;\; (288-88)/2 \;\; 88 \;\; 36 \;\; 4 \;\; -2 \;\; 4 \;\; -4]$$

$$= [188 \;\; 100 \;\; 88 \;\; 36 \;\; 4 \;\; -2 \;\; 4 \;\; -4]$$

$$[([188\;100]+[88\;36])/2 \;\; ([188\;100]-[88\;36])/2$$

$$4 \;\; -2 \;\; 4 \;\; -4] = [[138 \;\; 68] \;\; [50 \;\; 32] \;\; 4 \;\; -2 \;\; 4 \;\; -4]$$

$$= [138 \;\; 50 \;\; 68 \;\; 32 \;\; 4 \;\; -2 \;\; 4 \;\; -4 \;]$$

$$[([138 \;\; 50 \;\; 68 \;\; 32]+[4 \;\; -2 \;\; 4 \;\; -4 \;])/2$$

$$([138 \;\; 50 \;\; 68 \;\; 32]-[4 \;\; -2 \;\; 4 \;\; -4 \;])/2]$$

$$= [[71 \;\; 24 \;\; 36 \;\; 14] \; [67 \;\; 26 \;\; 32 \;\; 18]]$$

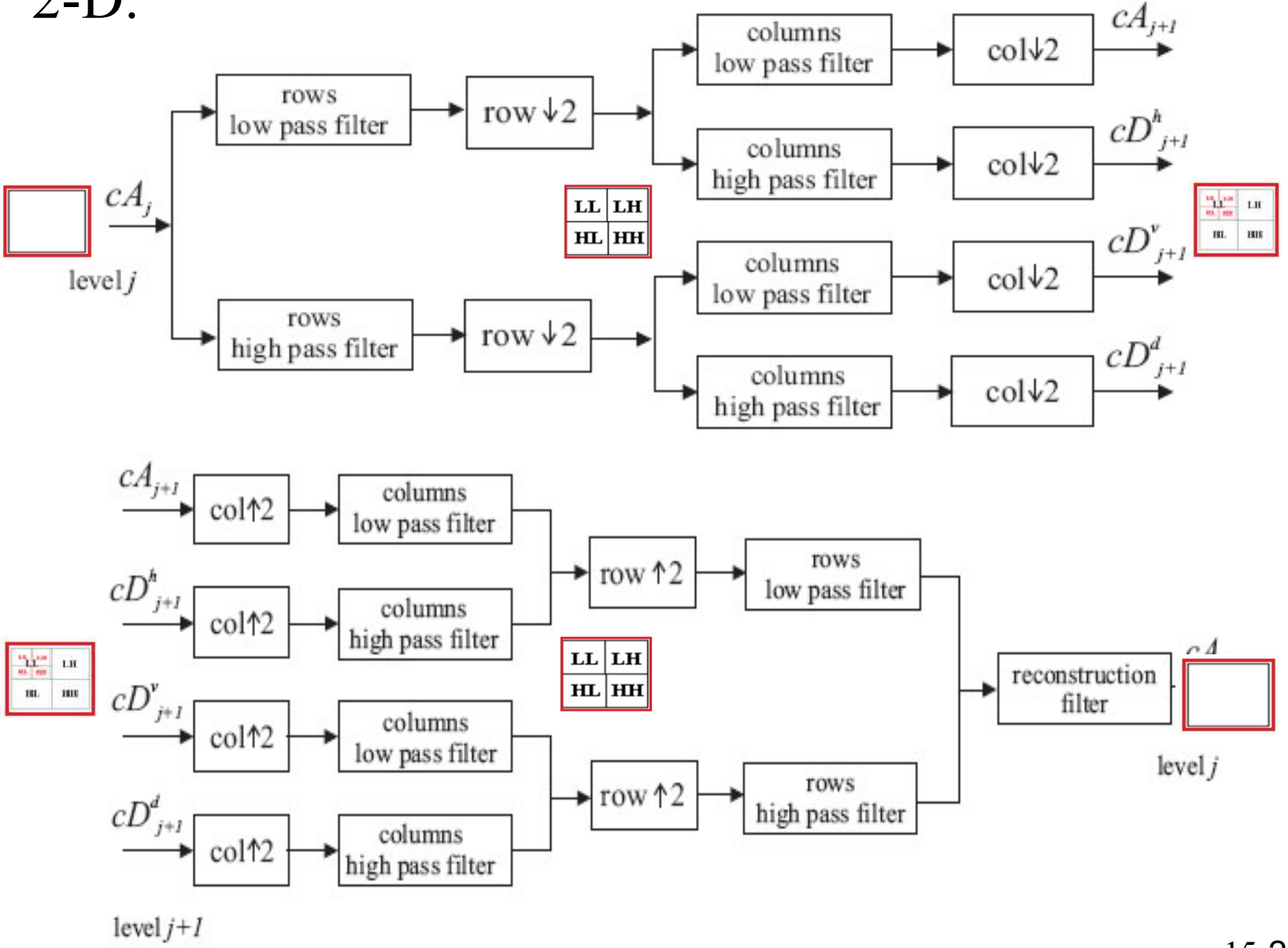
$$= [71 \;\; 67 \;\; 24 \;\; 26 \;\; 36 \;\; 32 \;\; 14 \;\; 18]$$

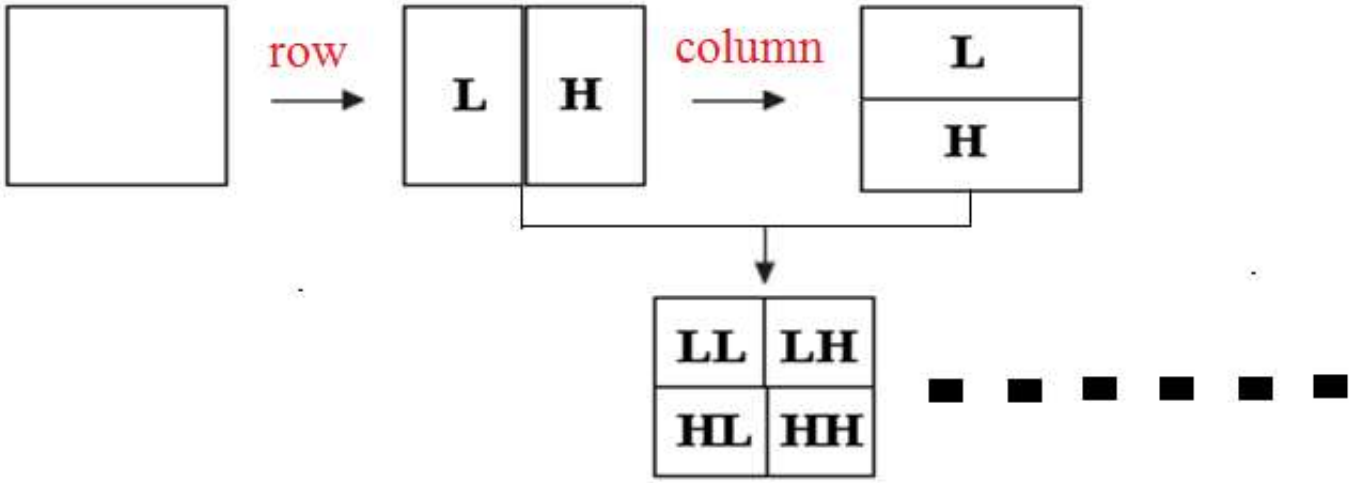
- If [4 –2 4 –4] are small, this leads to compression by letting the transform values [288 88 88 36 0 0 0 0]. Inverse transform results [69 69 25 25 34 34 16 16] close to [71 67 24 26 36 32 14 18].
- Remove negative values [4 0 4 0] leading to the transform values [288 88 88 36 4 0 4 0]. Its inverse transform results [71 67 25 25 36 32 16 16].
- Different filters can be used, e.g.,

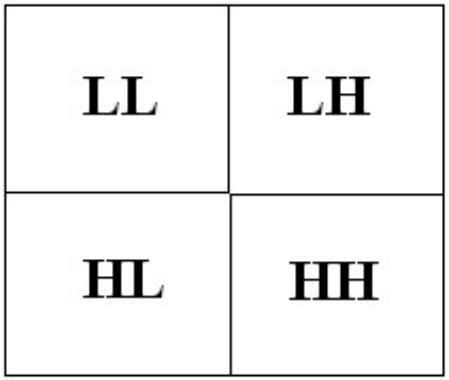
  Average ($s = (a + b) / 2$) as the low pass filter,

  Difference ($d = a - s$) as the high pass filter.

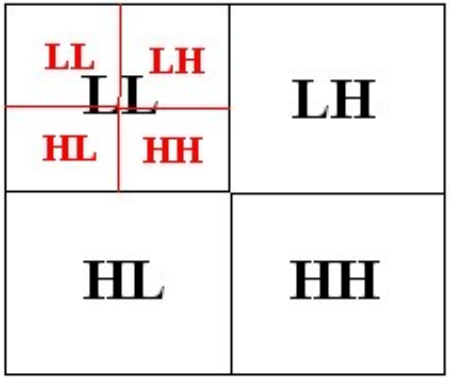
2-D:


cA_j
level j
rows low pass filter
rows high pass filter
row ↓2
row ↓2
columns low pass filter
columns high pass filter
columns low pass filter
columns high pass filter
col↓2
col↓2
col↓2
col↓2
cA_{j+1}
cD^h_{j+1}
cD^v_{j+1}
cD^d_{j+1}
LL LH HL HH
cA_{j+1}
cD^h_{j+1}
cD^v_{j+1}
cD^d_{j+1}
col↑2
col↑2
col↑2
col↑2
columns low pass filter
columns high pass filter
columns low pass filter
columns high pass filter
row ↑2
row ↑2
rows low pass filter
rows high pass filter
LL LH HL HH
reconstruction filter
level j
level $j+1$

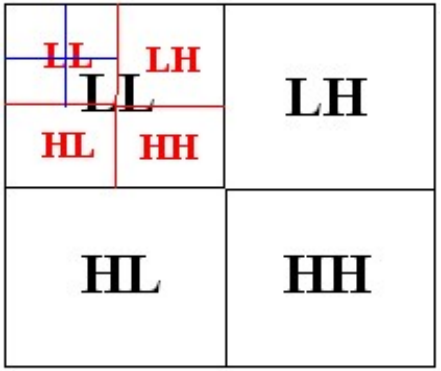




1-scale transform



2-scale transform



3-scale transform

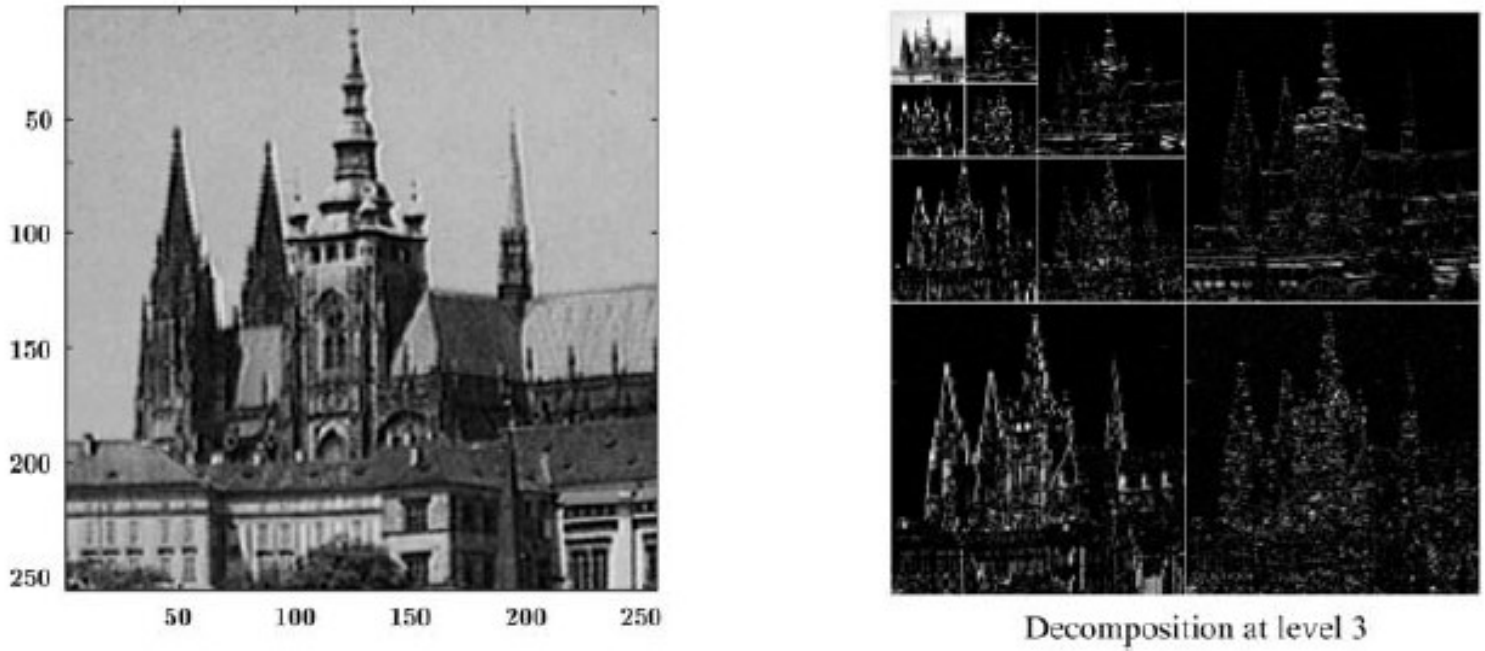


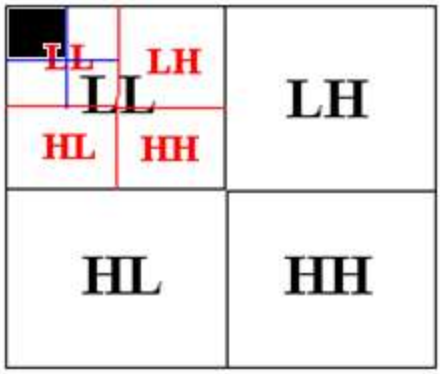
Figure 3.21: Decomposition by the 2D discrete wavelet transform. The Haar wavelet was used, and decomposition to three levels performed. On the left is the original 256×256 gray-scale image, and on the right four quadrants. The undivided southwestern, southeastern and northeastern quadrants correspond to detailed coefficients of level 1 at resolution 128×128 in vertical, diagonal and horizontal directions, respectively. The northwestern quadrant displays the same structure for level 2 at resolution 64×64 pixels. The northwestern quadrant of level 2 shows the same structure at level 3 at resolution 32×32 pixels. The lighter intensity 32×32 image in top left corner corresponds to approximation coefficients at level 3.

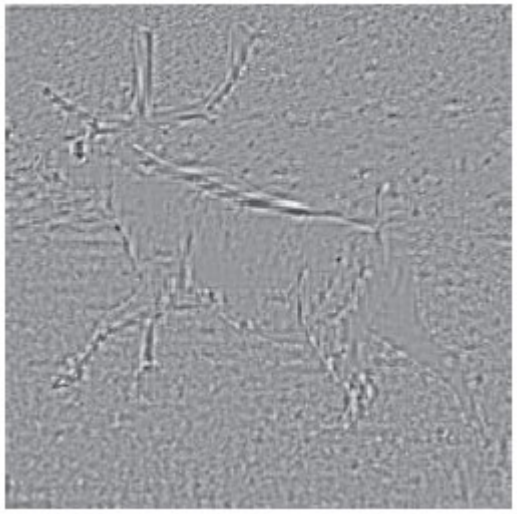
Example: High pass filtering

Input image

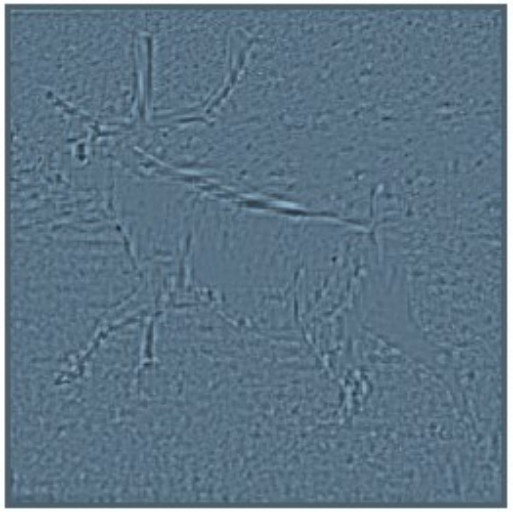


Set all the values to 0





2-scale decomposition



3-scale decomposition

Example: Denoising

Set DWT values that are smaller than d to 0.



Original image



Input noisy image



$d = 0.1$



$d = 0.3$



$d = 0.5$